

DBMS TASK – 5

Project Name: Payment Transaction Management System

Objective: To create a database for storing and managing customer payment transactions.

Database: PaymentDB

```
CREATE DATABASE PaymentDB;  
USE PaymentDB;
```

1. Create Payment Table & Store Details

Create a Payment table with payment mode, date, amount and status.

```
CREATE TABLE Payment (  
  Payment_ID INT PRIMARY KEY,  
  Customer_Name VARCHAR(50),  
  Payment_Mode VARCHAR(20),  
  Payment_Date DATE,  
  Amount DECIMAL(10,2),  
  Payment_Status VARCHAR(20)  
);  
  
INSERT INTO Payment VALUES  
(101, 'Yuvasree', 'UPI', '2026-08-25', 500.00, 'Successful'),  
(102, 'Anu', 'Card', '2026-08-25', 750.00, 'Successful'),  
(103, 'Priya', 'Cash', '2026-08-26', 300.00, 'Successful'),  
(104, 'Divya', 'UPI', '2026-08-26', 450.00, 'Failed'),  
(105, 'Kavi', 'Card', '2026-08-27', 900.00, 'Successful');
```

View the stored records:

```
SELECT * FROM Payment;
```

2. Manage Successful & Failed Transactions

The payment status is used to separate successful and failed transactions.

Successful Transactions

```
SELECT * FROM Payment  
WHERE Payment_Status = 'Successful';
```

Failed Transactions

```
SELECT * FROM Payment  
WHERE Payment_Status = 'Failed';
```

This helps the system identify completed and unsuccessful payments.

3. Analyze Payment Methods

COUNT() and GROUP BY are used to find how many times each payment method was used.

```
SELECT Payment_Mode, COUNT(*) AS Total_Transactions
FROM Payment
GROUP BY Payment_Mode;
```

Find the total amount received:

```
SELECT SUM(Amount) AS Total_Amount
FROM Payment;
```

Find the successful payment amount:

```
SELECT SUM(Amount) AS Successful_Amount
FROM Payment
WHERE Payment_Status = 'Successful';
```

4. Generate Payment Transaction Report

The following query generates a simple payment transaction report.

```
SELECT Payment_ID, Customer_Name, Payment_Mode,  
Payment_Date, Amount, Payment_Status  
FROM Payment  
ORDER BY Payment_Date;
```

Conclusion: The Payment Transaction Management System stores payment details, manages successful and failed transactions, analyzes payment methods, and generates useful transaction reports.

Tools Used: MySQL / SQL